

## Reviewer 1

### Comments

Reviewer 1 - Initial submission, 28/07/2026

This paper examines an LLM-hypothesized Fe-Mn-Al-Ni-Si-C composition and benchmarks its mechanical response against an established super-elastic Fe-Mn-Al-Ni alloy, employing synchrotron X-ray diffraction to rationalize the absence of super-elasticity. The topic is highly timely and carries significant implications for the growing field of AI-assisted alloy design, as it provides a rare experimental test of computationally generated hypotheses. The authors have collected a substantial dataset, and the finding that the AI-proposed alloy fails due to phase-stability mismatch rather than processing inadequacy is informative. However, despite the merits of this work, several aspects require systematic revision to strengthen the manuscript's clarity, logical flow, and evidential support. I believe that the paper is suitable for publication in this journal after major revisions. The specific revision comments are as follows:

1. The AI-designed alloy differs from the benchmark Omori alloy in multiple elemental concentrations simultaneously (Mn, Al, Ni, Si, and C are all varied). The authors attribute the failure of the AI-hypothesized composition primarily to its elevated carbon content and its destabilizing effect on the BCC parent phase. However, because this is a multi-variable compositional change, the specific role of carbon cannot be isolated. Without a designed control experiment—such as an AI-designed alloy variant without the 1000 ppm C addition—the assertion that "carbon is the main culprit" (as stated in the abstract and Section 4.1) remains speculative and insufficiently supported. The authors should either provide such a control or substantially temper their causal attribution, reframing the conclusion as a combined effect of the entire compositional modification.
2. The discussion of why the AI model failed (Section 4.3) is descriptive rather than diagnostic. The authors note that the model relied on thermodynamic and literature correlations, but they do not examine the model's specific reasoning pathway that led to the proposed composition. Did the model predict a BCC single-phase microstructure? Did it account for the formation of the D03 Fe<sub>3</sub>Al phase? A retrospective analysis using CALPHAD thermodynamic calculations (even a simple equilibrium phase diagram calculation for both alloys) would add significant quantitative rigor to the claim that the AI-designed alloy has insufficient chemical driving force for stress-induced transformation. Without such analysis, the "lessons learned" for AI-guided design remain vague.
3. The microstructural interpretation of the AI-designed alloy's deformation mechanisms lacks mechanistic depth. The authors assert that the absence of super-elasticity is due to "irreversible dislocation slip" rather than phase transformation, but they do not explain why this duplex FCC/BCC structure (62%  $\gamma$ , 34%  $\alpha$ ) preferentially deforms by slip. Is it the morphology of the two phases (e.g., lamellar or blocky) that constrains transformation? Or does the D03 Fe<sub>3</sub>Al phase pin dislocations in a way that suppresses the martensitic start stress? The authors should provide a more detailed mechanistic rationale linking the observed two-phase microstructure to the suppression of the transformation pathway.
4. The manuscript claims that the AI-designed alloy did not exhibit abnormal grain growth (AGG) even under the cyclic heat treatment intended to produce a bamboo-like structure. This observation, however, is buried in a single sentence in Section 3.3. Given that AGG is a central requirement for achieving super-elasticity in this alloy family, the fact that the AI-designed alloy fails to undergo AGG is a second, independent mode of failure. The authors should elevate this point, explicitly present the microstructure after

the AGG treatment (which is currently absent from the figures), and discuss how the multi-phase stability prevents grain boundary migration—this would strengthen the case that the AI design is fundamentally incompatible with the processing route.

5. The terminology used throughout the manuscript—specifically “Al-alloy” and “Al-hypothesized”—is problematic because “Al” is also the chemical symbol for aluminum, which is a major constituent of the alloy under study (~6 wt.%). This ambiguity is confusing and detracts from readability. The authors should replace “Al” with “AI” (artificial intelligence) or “AI-designed” in all instances, including the title, abstract, and keywords, to clearly distinguish the alloy designation from elemental aluminum.

6. In Section 2.1, the authors state that Si was added for solid-solution strengthening and to influence stacking-fault energy. However, this rationale is not revisited in the Discussion. The authors should either provide a more detailed justification for the Si addition or acknowledge in hindsight that Si may have contributed to the phase-stability imbalance.

7. The mechanical testing section (Section 2.4) describes the heating stage for the shape-memory effect test as “to  $\approx 200^\circ\text{C}$ ,” but the heating rate and holding time are not specified. These details are critical for assessing whether the reverse transformation was fully triggered. The authors should provide this information.

8. In Table 1, the Omori-alloy contains an unexpected 0.06 wt.% phosphorus, and the authors note the source is unknown. While this may not affect the main conclusions, the authors should discuss whether this unintentional impurity could have influenced the mechanical behavior of the benchmark alloy (e.g., by grain-boundary embrittlement) and whether it complicates the direct comparison with the AI-designed alloy.

9. The peak elongation of the AI-designed alloy ( $\approx 33\%$  at  $1000^\circ\text{C}$ ) is remarkably high for a duplex alloy. While this is not central to the main conclusion, the authors could briefly comment on whether this ductility is exceptional and whether it suggests any potential for structural applications despite the lack of shape-memory functionality.

Author - Initial submission, 21/08/2026

We thank the Reviewer for a careful reading, and in particular for the two requests that most improved this manuscript: the retrospective CALPHAD analysis (comment 2) and the promotion of the abnormal-grain-growth result to a full section with figures (comment 4). Section and figure numbers below refer to the revised manuscript, which adds Secs. 2.6, 3.4 and 3.5, Figs. 7–9 and the post-AGG micrograph as Fig. 1d; the discussion cited as Sec. 4.3 is now Sec. 4.4.

Comment 1.

Response. Agreed; the revision supplies the control and tempers the claim to it. Sec. 2.6 defines the control the Reviewer describes — the measured chemistry with carbon deleted — and Sec. 3.4 evaluates it beside both real alloys in three independently assessed databases, validated against the benchmark: with carbon the single-phase  $\alpha$  field opens only at  $\approx 1340^\circ\text{C}$ , above both the oxidation limit and calculated incipient melting; without it, at  $\approx 1150^\circ\text{C}$  (Table 3, Fig. 9c). A nickel scan from 4.2 to 7.8 at% does not move the solvus at all, so the benchmark’s nickel content would not have opened the window either. The claim is now the narrow one the control supports — carbon closes the  $\alpha$  field at the solution-treatment temperature — with the limits in Sec. 4.2 and the carbon-free heat named as the direct experimental test.

Comment 2.

Response. Both parts are done. Secs. 2.6 and 3.4, Fig. 9 and Table 3 give equilibrium step diagrams for both alloys in three databases, and they sharpened the diagnosis in the direction the Reviewer’s wording anticipates: not a BCC parent with insufficient driving

force, but no accessible heat treatment that produces that parent at all. On the reasoning pathway, the agent's 41-page report survives and is supplied as Supplementary Material, so both questions are answered from the document (Sec. 4.1): it did specify a BCC parent with coherent  $\beta$  precipitates and a 5–8% recoverable-strain target, but calculated no phase diagram and identified no solution-treatment window; and  $\text{D0}_3$  and  $\text{Fe}_3\text{Al}$  appear nowhere in it, though it discusses  $\kappa$ -carbide ordering at length for a different candidate family. Sec. 4.4 is rewritten around what these give together — the only occurrence of CALPHAD in forty-one pages is the report's own recommendation for future work.

#### Comment 3.

Response. Sec. 4.1 is substantially rewritten, and the calculations changed what the honest answer is. The parent is largely missing rather than reluctant: here the parent is BCC  $\alpha$  and the martensite FCC  $\gamma$  — the reverse of the carbon-steel convention, now stated in Sec. 3.3 — so a structure that is  $\approx 62\%$   $\gamma$  is mostly product phase already, and no accessible temperature produces single-phase  $\alpha$ . The  $\gamma$  excess also points to a transformation that has already run, since calculation gives  $\alpha$ -majority at 1200 °C while the quenched wire refines  $\gamma$ -majority, placing the martensite start above room temperature; calorimetry is named as the confirming experiment. On the Reviewer's two alternatives: SEM/EDS (Fig. 1b) shows an interlocked duplex in which transforming the discrete  $\alpha$  requires the majority  $\gamma$  to accommodate the shape strain, and  $\gamma$  yields at  $\approx 520$ – $675$  MPa, so slip preempts it; and  $\text{D0}_3$  at  $\approx 4\%$  is a bulk constituent, not a coherent dispersion, while coherent B2 is unavailable for the more basic reason that it requires quenching from a single-phase  $\alpha$  field — which also accounts for the failure of the 200 °C/3 h age (Fig. 2h).

#### Comment 4.

Response. Agreed, and done: one sentence with no figure has become Sec. 3.5, with the microstructure now shown twice — Fig. 1d, a fine banded duplex across the whole section with no grain approaching the section width, and Fig. 8, whose resolution into discrete spots is the grain-size evidence in reciprocal space. On mechanism, growth is not absent but arrested, and arrested at the second phase: a migrating boundary must drag or bypass a phase that no further time at temperature will remove, because the duplex state is the equilibrium state at 1200 °C (Sec. 3.4). The benchmark, single-phase  $\alpha$  there, has nothing to pin a boundary; Sec. 3.5 adds two reinforcing mechanisms of the same origin, transformation interfaces created on every cycle and solute drag from the partitioning each crossing entails.

We would respectfully qualify the framing on two points. The revision no longer says the alloy “did not exhibit AGG” — it does coarsen; what it does not form is bamboo, and Sec. 3.5 separates grain size from grain span throughout. And we do not read this as a second, independent failure mode: bamboo does not confer a transformation, it removes the constraint that limits one, as our own data show — the non-bamboo benchmark of Fig. 3b still recovers  $\approx 0.5\%$  on heating, an order of magnitude below the structure that optimises it, while the LLM-alloy shows nothing in any condition at any grain size. There is nothing here for bamboo to multiply, and coarsening embrittles this alloy besides (elongation 11.0%  $\rightarrow$  1.4%), so the route would cost its ductility first. Both failures trace to the same duplex constitution: one diagnosis, not two.

#### Comment 5.

Response. We have gone further than the letter of the comment: the test alloy is now the LLM-alloy and the reference the benchmark alloy throughout, with “AI-guided” reserved for the field in general, as in the keywords.

Comment 6.

Response. Correct that silicon was introduced in the Methods and never revisited; Sec. 4.2 now returns to it. What silicon did not do: as a ferrite stabilizer it cannot have caused the  $\gamma$ -stability imbalance, the carbon-free control holds Si at its measured level and still returns 100%  $\alpha$  at 1200 °C, and at half its intended level it cannot have delivered the strengthening it was specified for either. What it may have done is the hindsight acknowledgement asked for: Heo et al. (Metall. Mater. Trans. A 43 (2012) 1731) found Si promoting  $(\text{Fe,Mn})_5(\text{Si,Al})\text{C}$  together with  $\text{D0}_3$  ordering in the ferrite —  $\text{D0}_3$  being the ordered phase we observe — so an addition the design treated purely as a strengthener has a documented second role as a promoter of the very ordering product found, which the reasoning that proposed it did not engage with. With two heats we cannot separate a silicon contribution to that ordering from an iron-aluminum one, and do not claim it.

Comment 7.

Response. Added to Sec. 2.4: heated to  $\approx 200$  °C at  $\approx 50$  °C s<sup>-1</sup>, held  $\approx 10$  s. At 0.36 mm the section reaches temperature in a fraction of that hold. Sec. 2.4 also now states that the specimen was unloaded to zero stress and the crosshead returned to zero strain before heating. The same protocol resolved 0.5% recovery in the benchmark (Fig. 3b) — the internal evidence that the heating stage sufficed where retained martensite existed.

Comment 8.

Response. Sec. 2.1 now discusses this. At 0.06 wt% ( $\approx 0.10$  at%) the phosphorus is below the level at which grain-boundary embrittlement is normally reported in Fe-Mn austenitic and duplex alloys, and the benchmark showed both good ductility and a clear stress-induced transformation. Any effect it did have could only have degraded the benchmark, and the comparison runs “benchmark transforms, LLM-alloy does not” — so the impurity makes that comparison conservative rather than confounded. A residual from the commercial melting stock is the most plausible origin, stated as unconfirmed.

Comment 9.

Response. Added at the end of Sec. 4.3, and the honest answer — respectable rather than exceptional — is itself informative. The Ni-alloyed Fe-Mn-Al-C low-density steels whose compositional field this alloy occupies span  $\sigma_{0.2} \approx 500\text{--}940$  MPa, UTS  $\approx 710\text{--}1020$  MPa and elongation 8–78% (Rahnama et al., Acta Mater. 132 (2017) 627), and the 1000–1200 °C conditions of Table 2 sit inside that envelope on all three measures. The revision therefore frames the alloy as the question suggests: a serviceable structural metal that was hypothesized to be a shape-memory alloy and is not one — an unoptimized member of the low-density-steel family rather than a failed member of the shape-memory family.

We thank the Reviewer again: the two central requests — the thermodynamic calculation and the elevation of the AGG result — did the most of any comments we received to strengthen the manuscript.

Reviewer 1 - Revision 1, 25/08/2026

No comment.

## Reviewer 2

### Comments

Reviewer 2 - Initial submission, 31/07/2026

I go through the paper and my advising is mentioned in following.

The authors investigate “Fe-based shape memory alloy; super-elasticity; AI-guided alloy design; synchrotron XRD; abnormal grain growth”. The structure of paper was arranged well and nicely written, but can find some mistake in this paper. The authors are advised to carefully read through the manuscript and correct these mistakes. Abstract clearly summarizes. Experimental procedure clearly described. Result was presented well, but there are few weaknesses in discussion part. Conclusions appropriate in view of the study objectives and results. However, although the study has been carried out accordingly, there are still errors and issues required to be rectified and listed as follow:

Title:

- 1- Suggestion and consider specifying "LLM-hypothesized" instead of the broader term "AI-hypothesized" for greater precision.
- 2- Adding the phrase "experimental validation" would better reflect the manuscript's main contribution.
- 3- The title is technically appropriate but could highlight the role of phase stability in the observed mechanical response.

Abstract:

- 1- The novelty of experimentally validating an LLM-generated alloy should be stated more explicitly.
- 2- Quantitative mechanical results (e.g., recoverable strain or yield strength) should be included to support the conclusions.
- 3- The influence of carbon on phase stability should be briefly justified rather than only suggested.
- 4- The abstract should briefly explain why the benchmark alloy exhibits super-elasticity while the AI alloy does not.

Introduction:

- 1- The literature review should include more recent studies on AI-driven alloy design beyond LLM applications.
- 2- The scientific gap between computational alloy prediction and experimental validation should be better defined.
- 3- The rationale for selecting this specific alloy composition should be explained more thoroughly.
- 4- The research objectives and hypotheses should be clearly stated at the end of the introduction.

Experimental procedure:

- 1- The alloy design workflow generated by the LLM should be described in sufficient detail to ensure reproducibility.
- 2- Standard No. of tensile test must be mentioned.
- 3- Statistical repeatability of the mechanical tests should be reported.

Results and Discussion:

- 1- The discussion should quantitatively correlate phase fractions with the observed mechanical behaviour.
- 2- The proposed role of carbon in suppressing super-elasticity should be supported by thermodynamic calculations.
- 3- Fig. 2 is small and unclearly.

- 4- Fig. 2 e, should be presented and discussed.
- 5- The implications of this negative result for future AI-assisted alloy design deserve deeper discussion.
- 6- Is there any formation Martensite phase in microstructure, this should be present and discuss.

#### Conclusion:

- 1- The conclusions should distinguish experimental observations from the authors' interpretations.
- 2- Emphasize the broader implications for AI-assisted materials discovery rather than this alloy alone.
- 3- Discuss the limitations of relying solely on LLM-generated compositions.
- 4- Include recommendations for integrating thermodynamic screening into AI-guided alloy design.

Therefore, in the submitted paper, the above mentioned comments will make it clear that the manuscript can be accepted for its publication, but required to revise.

**Attached file:** Comments list-Fe-Mn-Al-Ni-Si-C Alloy -Author.doc

Author - Initial submission, 21/08/2026

We thank the Reviewer for a thorough, well-organized review. All twenty-four points have been acted on, and three of them are now the title of the paper. Section and figure numbers below refer to the revised manuscript; the revision adds Secs. 2.6, 3.4 and 3.5 and three figures (Figs. 7–9), so some numbering has shifted.

#### Title

Response (1–3). All three are adopted together: Experimental Validation of an LLM-Hypothesized Fe-Mn-Al-Ni-Si-C Alloy: Phase Stability Governs the Absence of Super-elasticity. This also repairs a defect in the submitted title, which could be mis-parsed as claiming the alloy is super-elastic. The test alloy is renamed the “LLM-alloy” throughout and the reference alloy the “benchmark alloy”; “AI-guided alloy design” is kept as a keyword, being the accepted name of the field.

#### Abstract

Response (1-2). Added on both sides of the comparison: 0.2% proof stress from 1908 to 519 MPa across the anneal series and elongation peaking at 33%; strain recovery indistinguishable from elastic springback against  $\approx 0.5\%$  recoverable transformation strain in the benchmark; single-phase  $\alpha$  at  $\approx 1340^\circ\text{C}$  versus  $\approx 1150^\circ\text{C}$  once carbon is deleted, and a  $\approx 0.10$  wt% carbon threshold against 0.010 wt% in the benchmark.

Response (3). In the submitted abstract the carbon claim was asserted; it is now justified by calculation. The abstract gives the mechanism and its basis: equilibrium CALPHAD across three independent databases, validated against the benchmark, shows the benchmark solution-treating to single-phase BCC  $\alpha$  at  $1200^\circ\text{C}$  while the LLM-alloy is two-phase there and reaches single-phase  $\alpha$  only at  $\approx 1340^\circ\text{C}$ , and deleting carbon alone lowers that field to  $\approx 1150^\circ\text{C}$ . Full analysis in Secs. 2.6 and 3.4 (Fig. 9, Table 3), with the causal claim elsewhere tempered to what the calculation supports.

Response (4). The abstract now explains it through phase stability: the benchmark

reaches the single-phase BCC  $\alpha$  parent at 1200 °C; the LLM-alloy cannot reach it within its processing range, so the parent state on which stress-induced transformation depends is unavailable. The role of the coherent B2 precipitates within that parent is developed in Secs. 3.4 and 4.1.

## Introduction

Response (1). The opening now surveys the family: property-targeted screening over databases and surrogate models (the AIMS framework for shape memory alloys), generative inverse design, and LLM-based hypothesis generation, alongside the GNoME-scale prediction work already cited. It closes on what they share — a composition that has never been made.

Response (2). A dedicated gap paragraph now follows (Sec. 1): generating candidates is cheap, testing one is not and is correspondingly rare, and publishing a candidate that failed is rarer still. It also names the specific technical gap — models that optimize thermodynamic stability need not check that the required phase fields are reachable by any route the laboratory can run.

Response (3). Treated at three levels. Sec. 1 states the agent’s hypothesis in its own terms (reduced Ni for cost, Si to stabilize the  $\alpha$  parent,  $\approx 1000$  ppm C for interstitial strengthening, no bamboo structure required). Sec. 2.1 documents its provenance — Hypothesis A2, ranked joint second of six families, its composition window, and the fact that the melt lies outside that window in Mn and Al, a deviation then tested by calculation in Sec. 3.4. Sec. 2.1 also places the composition in the literature it actually sits in, the Ni-alloyed Fe-Mn-Al-C low-density steels.

Response (4). Done. The Introduction now ends on the hypothesis paragraph and an explicit objective: “to determine not merely whether that hypothesis holds but where precisely it fails: Secs. 3.1–3.3 establish what the alloy did, Secs. 3.4 and 4.2 establish why, and Sec. 4.4 asks what a design workflow would have had to do differently.” This replaces a closing sentence another reviewer identified as overclaiming.

## Experimental procedure

Response (1). Sec. 2.1 now gives the agent and configuration (Gemini Deep Research), the report’s compilation date (4 June 2025), its sources (76, accessed 19 May 2025), its six candidate families and their ranking, the composition window of the hypothesis selected, and its performance target (5–8% recoverable strain at 400–700 MPa). The entire 41-page report is supplied as Supplementary Material, so the reader has the generative artifact itself. One element is unrecoverable and the revision says so: the verbatim prompt was not preserved and the report does not embed it.

Response (2). Sec. 2.4 now states this. Specimen handling and the monotonic-tension elements broadly follow ASTM E8/E8M; the cyclic protocol is not a standardized test but a purpose-designed method for fine superelastic wire, cited to the publication describing it on the same apparatus and gauge (Cai et al., Shape Mem. Superelasticity 10 (2024) 460–472). ASTM F2516 is deliberately not cited: it prescribes a different cycle for NiTi and would misdescribe what was done. Sec. 2.4 also gives the parameters — displacement control, 127 mm gauge,  $0.25 \text{ in min}^{-1}$  ( $8.3 \times 10^{-4} \text{ s}^{-1}$ ), strain from crosshead extension without extensometry, so the moduli are apparent rather than true.

Response (3). Based on the Reviewer’s comment, we carried out additional heat treatments and tensile testing so that every condition is now replicated. Table 2 is

updated: each entry is the mean of three specimens with one standard deviation, and the caption states that each data point is the average of three samples (Sec. 2.4 reports the same). The replicates confirm the sweep rather than change it — every ranking is preserved and the largest relative standard deviation is  $\approx 6.5\%$ , against a  $\approx 3.7\times$  range in strength and  $\approx 16\times$  in elongation. Where scatter is itself informative it is reported as such: it is large in the abnormal-grain-growth condition, and Sec. 3.5 reads it as evidence of boundary cracking.

## Results and Discussion

Response (1). Sec. 4.1 now makes the correlation explicit in three steps. A structure that is  $\approx 62\%$   $\gamma$  is mostly the product phase of this family's transformation (parent BCC  $\alpha \rightarrow$  martensite FCC  $\gamma$ , now spelled out in Sec. 3.3), leaving the  $\approx 34\%$   $\alpha$  as the only parent available. The calculated 1200 °C equilibrium (62–71%  $\alpha$ , Table 3), inverted against the measured quenched state, indicates a partial  $\alpha \rightarrow \gamma$  transformation ran on cooling. The yield strengths of Table 2 ( $\approx 520$ – $675$  MPa) then set the stress at which slip in the majority  $\gamma$  preempts transformation of the  $\alpha$  it surrounds. The  $\approx 4\%$   $D0_3$  correlates negatively: as a bulk constituent rather than a coherent dispersion it supplies none of the slip resistance nanoscale B2 supplies in the benchmark.

Response (2). This is the largest single addition to the revision. New Secs. 2.6 and 3.4, with Fig. 9 and Table 3, give equilibrium CALPHAD step diagrams for both alloys from 400 to 1400 °C, computed from the measured chemistries in three independently assessed open databases and validated against the benchmark, whose single-phase  $\alpha$  field and  $\approx 20\%$  ordered bcc they reproduce without fitting. The carbon result comes from a virtual control — the measured chemistry with carbon deleted — now plotted as Fig. 9c beside the measured composition so the restored  $\alpha$  field can be read directly. All three databases agree:  $\approx 0.10$  wt% carbon moves the  $\alpha$  solvus by  $\approx 190$  °C, the difference between a solution treatment that cannot be run and one that can. A nickel scan separates the other suspect element cleanly — nickel does not move the solvus; it controls the amount of ordered bcc.

Response (3). Fig. 2 has been rebuilt from the raw Instron records rather than rescaled: eight panels at 600 dpi at column width, uniform axes, legible ticks (the submitted version clipped “2.5” to “2.”). The caption now states what stress and strain are computed from, including the measured per-spool diameters and the 127 mm gauge.

Comment 4. Fig. 2 e, should be presented and discussed.

Response (4). Done, and the defect was broader: the submitted text referred to no Fig. 2 panel by letter and showed six of the eight Table 2 conditions. The rebuilt figure presents all seven annealed conditions plus the aged one, (a)–(h), and Sec. 3.2 walks through them, including the panel named — 1000 °C, where yielding falls to  $\approx 675$  MPa and the series reaches its elongation peak. The narration closes on the feature common to all eight: every unloading segment is straight, parallel to the elastic line, and returns no strain beyond  $\sigma/E$ .

Response (5). Sec. 4.4 is substantially expanded, and the new calculations turn a general caution into a concrete prescription: an equilibrium step diagram evaluated not for whether the desired phase exists but for whether the phase fields the intended route requires overlap the temperatures at which that route can be run. It also records that the agent's own report recommended CALPHAD screening without performing it — placing the failure in the workflow's termination rather than the model's knowledge — and bounds the scope explicitly to one composition, one session, one route.



Response (6). The submitted manuscript never stated what “martensite” is in this family, whose convention is the reverse of that familiar from carbon steels; Sec. 3.3 now says so explicitly — the parent is BCC  $\alpha$  and the martensite formed from it is FCC  $\gamma$ , so an alloy rich in  $\gamma$  has less parent available to transform, not more. On the direct question: the refinement resolved  $\gamma$ ,  $\alpha$  and  $\text{D0}_3$   $\text{Fe}_3\text{Al}$  and nothing else, with no  $\epsilon$ -HCP or  $\alpha'$  needed to fit any pattern from either alloy, and the  $\gamma$  in the annealed LLM-alloy is thermally stable retained  $\gamma$ , unchanged by 10% deformation (Figs. 4a–b, 6a). Secs. 3.1 and 4.1 go one step further in the direction the question points: the lath-like morphology of Fig. 1d and the inversion between the calculated  $\alpha$ -majority equilibrium and the measured  $\gamma$ -majority state both suggest part of the  $\gamma$  formed thermally on cooling, i.e. the martensite start lies above room temperature. Calorimetry is named as the confirming experiment.

## Conclusion

Response (1). The Conclusions are restructured to enforce the separation: an observation paragraph (“What was observed is the following” — castability, the 1908 → 519 MPa sweep, the 33% elongation peak, no departure from elastic springback in any condition, the benchmark’s  $\approx 0.5\%$  recovery, the diffraction results), then a separate interpretation paragraph, then the calculation-based diagnosis with its evidential status stated, and finally three limitations.

Response (2). The Conclusions now state it as a transferable procedure rather than a sentiment: screen LLM-proposed compositions for competing phase stability and interstitial content before synthesis, by an equilibrium calculation checked against the intended route’s temperature window. We should be transparent that another referee pressed in the opposite direction, toward narrowing the paper to an alloy case study. The revision splits the claim: the experimental findings are bounded as a case study, while the methodological claim, resting on the diagnosis rather than the sample size, is retained and emphasized as the Reviewer asks.

Response (3). The limitations paragraph now names the specific limitation this case demonstrates: composition-only reasoning is blind to the distinction that decided it — not whether a phase is stable, but whether the temperature at which it becomes stable can be reached in the available equipment. An LLM reasoning from literature correlations has no reliable way to represent that constraint, and instructing it to consult thermodynamic data did not supply one: its report recommended CALPHAD screening without performing it.

Response (4). Adopted, and now demonstrated rather than merely recommended. Sec. 4.4 specifies the screen concretely — the two-part step-diagram test against the processing window, requiring no proprietary software and no measurement on the candidate itself. Sec. 4.2 shows it operating on this alloy, rejecting the composition where the intended 1200 °C treatment is found to lie  $\approx 140$  °C below the single-phase field. The abstract’s final sentence carries it in one line.

We thank the Reviewer again; the twenty-four comments were actionable, and the manuscript is measurably better for all of them.

Reviewer 2 - Revision 1, 03/09/2026

I have now reviewed the revised version of the above-mentioned manuscript, which was resubmitted by the authors in response to my previous comments and those of the other reviewers.

I am pleased to state that the authors have comprehensively and satisfactorily addressed all the points raised in my initial review.

In my opinion, the current version represents a substantial improvement over the original submission. The study is now methodologically sound, the data support the conclusions drawn, and the findings are of sufficient novelty and interest to the readership of Journal of Materials Research and Technology.

Given that the authors have fully resolved all previous issues and that the manuscript now meets the journal's standards for quality and originality, I recommend acceptance of this paper in its current form.

## Reviewer 3

### Comments

Reviewer 3 - Initial submission, 28/07/2026

Thank you for submitting the manuscript entitled “Mechanical Responses of an AI-Hypothesized Super-elastic Fe-Mn-Al-Ni-Si-C Alloy”. This study experimentally evaluates an AI-assisted alloy candidate and reports its lack of superelasticity under the tested conditions. However, the current manuscript does not provide sufficient evidence to support its main comparisons and mechanistic conclusions. In particular, the effects of heat treatment, grain structure, texture, and nanoscale precipitates have not been adequately characterized. Therefore, I do not recommend publication of the manuscript in its present form. The specific reasons are listed below:

1. The statement at the end of the Introduction that “because the processing history was kept identical for both alloys, compositional differences are the sole variable determining the final mechanical properties” is not valid from a materials-science perspective. It is not meaningful to compare the superelastic differences between the AI alloy and the Omori alloy solely by applying the same processing route, because identical external processing parameters do not mean that the two compositions achieve the same degree of recrystallization, grain size, grain orientation, texture, phase constitution, dislocation density, and precipitation state. The manuscript may state that the two alloys were subjected to the same nominal processing parameters, but further justification is required before the final performance differences can be attributed solely to chemical composition.
2. The basis for designing the candidate composition in Section 2.1 is questionable. Fe-Mn-Al-C, Fe-Mn-Al-Ni-C, and Fe-Mn-Al-Si-C systems with compositions similar to that studied here have mainly been investigated in the literature as low-density or lightweight structural steels. Their B2, DO3, and carbide phases are generally used to improve strength and ductility rather than to produce a reversible martensitic transformation. For example, Rahnama et al. investigated the effect of the B2 phase on the microstructure and mechanical properties of Ni-alloyed Fe-Mn-Al-C lightweight steels (*Acta Materialia* 132 (2017) 627–643); Saha et al. reported that NiAl-type nanoscale B2 precipitates are mainly located in the BCC regions of low-density Fe-Mn-Al-C steels (*JOM* 74 (2022) 3181–3190); and Heo et al. showed that Si can promote the formation of DO3 ordered phases and complex carbides in Fe-Mn-Al-C steels (*Metallurgical and Materials Transactions A* 43 (2012) 1731–1735). The authors should clearly explain why this alloy composition was selected for investigation and, given that many similar compositions mainly exhibit the microstructures and deformation behavior of lightweight steels, why this composition was expected to show room-temperature superelasticity.
3. Table 1 shows that the actual Si content of the AI alloy is 1.11 wt.%, whereas the nominal content is 2.2 wt.%; the actual value is approximately half of the designed value. Considering that Si is intended by the authors to strengthen the matrix and regulate phase

stability and precipitate formation, this deviation cannot be described as “very close to the nominal values.” Strictly speaking, the claimed nominal composition requires further verification in this study.

4. Section 2.2 describes a cyclic heat treatment for abnormal grain growth. However, the main text only briefly states that the XRD pattern of the AGG-treated sample is similar to Fig. 4a, without presenting the grain size, formation of bamboo-like grains, grain orientation, texture, or corresponding cyclic stress-strain results after this treatment. The key mechanical specimens in Figs. 2 and 3 were not in this cyclic heat-treated condition. The key heat treatment specifically included in the Methods does not enter the core evidence chain, and the reporting of the results is incomplete.

5. Section 2.2 also raises a question regarding the applicability of the processing route. The formation, size, and distribution of B2 nanoscale precipitates depend on the specific composition, aging temperature, cooling rate, and subsequent aging treatment. After the composition is changed to one containing Si and C with a lower Ni content, the optimum precipitation window may be entirely different. Directly applying the heat treatment used for the Omori alloy to the new composition can only show that the two alloys respond differently under the same treatment. Further study is required to determine whether this processing route is suitable for the new alloy and whether it can demonstrate that the new alloy itself cannot form an appropriate B2 nanoscale phase or exhibit superelasticity.

6. Section 2.5 uses ex-situ XRD before deformation and after unloading. For a superelastic material, stress-induced martensite may form during loading and reverse completely during unloading. Therefore, similar diffraction patterns before loading and after unloading do not necessarily exclude a reversible phase transformation during loading. More direct evidence should be provided to confirm the deformation mechanism.

7. Section 3.2 systematically presents only the cyclic curves of the AI alloy and does not provide fully corresponding stepwise cyclic loading-unloading curves for the Omori alloy. Many factors affect superelastic performance. The authors should ensure that the grain size, orientation, texture, and precipitation state of the two alloys are characterized on a corresponding basis before comparing their properties.

8. The identification of B2 and DO3 in Section 3.3 cannot support the nanoscale precipitation mechanism proposed in the manuscript. Whether the absence of an obvious B2 peak in the AI alloy proves the complete absence of B2 still requires further discussion. The authors should assess whether the heat-treatment route used for the AI alloy is appropriate and use additional characterization methods to determine whether the B2 phase is present.

9. Section 4.1 mainly attributes the absence of superelasticity to the stabilization of the FCC  $\gamma$  phase by C. However, the contents of Fe, Mn, Al, Ni, Si, and C all differ simultaneously between the AI alloy and the Omori alloy, and the reduction of Ni from 7.5 at.% to 4 at.% may also significantly affect B2 precipitation and parent-phase stability. How can the authors identify C as the main factor based on only two compositions in which multiple elements vary simultaneously?

10. The statement about B2 in Section 4.1 is too absolute. “Omori-alloy relies on coherent, ordered  $\beta$ -NiAl (B2) precipitates” may easily imply that B2 is the sole or necessary condition for superelasticity. Appropriate nanoscale B2 precipitates in conventional Fe-Mn-Al-Ni alloys do indeed help increase resistance to slip and improve transformation reversibility (Omori et al., *Applied Physics Letters* 101 (2012) 231907; La Roca et al., *Journal of Alloys and Compounds* 708 (2017) 422–427). However, superelasticity also depends on parent-phase stability, transformation temperature, grain size, grain orientation, texture, and the competition between transformation stress and plastic yield stress.

11. Section 4.2 interprets 600-700 °C as recovery, 800 °C as the onset of recrystallization,

and 800-1200 °C as grain growth after complete recrystallization. However, these interpretations are based mainly on changes in strength and elongation and lack corresponding microstructural and EBSD evidence at each temperature. The statements “from 800 to 1200 °C the wire is fully recrystallized” and “single-phase-controlled structural metal” should not be presented as definitive conclusions without direct evidence.

12. The discussion in Section 4.3 overgeneralizes the limitations of AI-assisted design. This study examines only one candidate selected by the researchers from a range proposed by the model. At most, the current results show that this specific candidate and the processing route used in this study did not achieve the target performance. The manuscript should be repositioned as a case study of the composition-processing-microstructure-property relationships of a specific Fe-Mn-Al-Ni-Si-C alloy, with AI treated only as the source of the candidate rather than as the basis for the main generalized conclusion of the manuscript. In summary, in summary, the current manuscript has not yet met the publication standards of this journal.

Author - Initial submission, 21/08/2026

We thank the Reviewer for its four themes — that “identical processing” does not license a composition-only comparison, that this composition sits in the lightweight-steel literature, that the AGG treatment did not enter the evidence chain, and that several mechanistic statements outran their evidence — are all correct, and the revision concedes each and rebuilds the affected sections. It also adds the analysis we believe answers the underlying objection: three-database equilibrium CALPHAD calculations (Secs. 2.6 and 3.4, Fig. 9, Table 3), validated against the benchmark. Section and figure numbers refer to the revised manuscript, in which Secs. 2.6, 3.4, 3.5 and Figs. 7–9 are new, so the Discussion sections cited as 4.2 and 4.3 are now 4.3 and 4.4; the alloys are renamed the “LLM-alloy” and the “benchmark alloy” throughout.

Comment 1.

Response. The Reviewer is right and the sentence is deleted; Sec. 1 now states the defensible form — the alloys were put through the same nominal route, which in two different compositions guarantees nothing about recrystallization, grain size, texture or precipitation state, so what follows compares how each responds to a common route rather than isolating composition. Two things supply the further justification asked for. Sec. 4.1 makes explicit the one respect in which the alloys were not treated identically, because it runs against the concern rather than with it — the LLM-alloy received the 200 °C/3 h age reported to develop pseudo-elasticity in this family and showed no response whatever (Fig. 2h), while the benchmark received no age at all and still showed reverse transformation on heating — and the attribution to composition no longer rests on the two-heat comparison at all, since Sec. 3.4 shows there is no single-phase  $\alpha$  field at any temperature this alloy’s processing can reach.

Comment 2.

Response. All three supplied references are now cited and used. Sec. 2.1 documents the composition’s provenance — Hypothesis A2 of the agent’s report, supplied in its 41-page entirety as Supplementary Material — and Sec. 1 states that hypothesis in the agent’s own terms so the reader sees what is under test.

On why it was expected to be superelastic, the honest answer is now given: the agent reasoned from the shape-memory literature and did not engage with the lightweight-steel literature its composition actually lands in. Sec. 2.1 states the overlap at first introduction — essentially the same Ni-Al pairing at essentially the same Ni level as Rahnama’s and

Saha's compositions, differing chiefly in about twice the manganese and an eighth of the carbon — and Sec. 4.2 develops the comparison: the two families form the same ordered phases but put them to opposite purposes, coherent B2 within a single-phase  $\alpha$  parent letting a transformation outrun plasticity in one, the same precipitates serving as strengtheners in a structure never intended to transform in the other. Judged against the second family this alloy is neither anomalous nor a failure; the failure lies in the expectation attached to it. What the Reviewer identified as a questionable design basis is, examined closely, the paper's clearest finding — the agent proposed a lightweight-steel composition while believing it was proposing a shape-memory alloy.

Comment 3.

Response. Correct, and the phrase is deleted. Sec. 2.1 now reports the deviation quantitatively — silicon recovered at  $\approx 50\%$  of intent, consistent with oxidation loss during vacuum induction melting — and the LLM-alloy is identified by its measured composition everywhere the nominal was previously used as its identity, with every calculation in Secs. 2.6/3.4 taking the measured chemistry as input, so no conclusion rests on the nominal. Sec. 4.2 states what the shortfall means: at half its intended level silicon cannot have delivered the strengthening it was specified for, while the carbon-free control restores the single-phase  $\alpha$  field regardless, so the silicon deficit did not close the processing window either. The benchmark alloy is close to nominal, so the caveat attaches to one alloy only, and the text says so.

Comment 4.

Response. Accepted, and this drove more new content: the AGG condition was one sentence and is now Sec. 3.5, with two new figure elements — Fig. 8 (post-AGG diffraction) and Fig. 1d (post-AGG optical microstructure). On the evidence chain, Sec. 3.5 reports the outcome for both alloys, connects it to Sec. 3.4 as its mechanism and feeds Secs. 4.1 and 4.4; we also correct the sentence quoted, since the post-AGG pattern is not “similar to Fig. 4a” — same phases, but the rings resolve into markedly more discrete spots, and Sec. 3.3 now reports phase and grain size separately.

On the Reviewer's remaining items: the benchmark bamboos completely, which Sec. 3.5 states without presenting it as a finding of this paper, while the LLM-alloy's response — the actual finding — is shown directly in Fig. 1d, a fine banded duplex across the entire section with no grain approaching the section width, and Sec. 3.5 distinguishes throughout between grain size, which the treatment does increase, and grain span, which it does not. Texture is now data (Fig. 7, inverse pole figures for both phases). The cyclic record is reported and is poor in an informative way: specimens fail near 1% elongation with large scatter and visible boundary cracking, so the treatment introduces cracking without producing the structure it was applied to produce; stepwise curves were not recorded because in grain size this condition lies between the 1- and 40-minute anneals at 1200 °C, both characterised and neither showing any transformation, and a response bracketed by two such curves cannot itself show one. Sec. 3.5 then draws the conclusion this supports — bamboo amplifies a transformation, it does not create one — so the AGG experiment closes a route rather than leaving one untested.

Comment 5.

Response. We concede the logical point without reservation: the experiments alone show only that two alloys respond differently to one route, and the revision says exactly that in Sec. 1 and again in Sec. 4.1, with route optimisation named as future work in the Reviewer's own terms.

The new analysis bears directly on the mechanism proposed — that the optimum

precipitation window may simply lie elsewhere — and its result is that the difficulty is not a misplaced window but a missing prerequisite. Coherent nanoscale B2 is obtained by quenching from a single-phase  $\alpha$  field and ageing below it, and the LLM-alloy has no accessible single-phase  $\alpha$  field at any temperature: its solvus is  $\approx 1340$  °C, above both the 1200 °C practical ceiling and the  $\approx 1240$  °C incipient-melting temperature, while the ordered bcc it can form at equilibrium is  $\leq 11\%$ , confined below  $\approx 850$  °C and sitting within a  $\gamma$  matrix. There is therefore no ageing temperature at which this composition could develop the benchmark's microstructure, because the step before ageing is unavailable everywhere — the window cannot be relocated to a temperature that does not exist. The experimental caveat stands, these being equilibrium calculations.

Comment 6.

Response. The Reviewer is right that ex-situ diffraction cannot exclude a fully reversible transformation, and the revision restructures the argument so the diffraction no longer carries that weight. Sec. 3.3 now states the order of evidence explicitly: a transformation reversible enough to leave no diffraction signature after unloading would still have had to produce recoverable strain during unloading, and the unloading segments of Fig. 2 are straight, parallel to the elastic line and return no strain beyond  $\sigma/E$  in all eight conditions; the loading–unloading–heating test of Fig. 3a closes the remaining route, since retained martensite stable enough to survive unloading would revert on heating, as it does for the benchmark and does not for the LLM-alloy. The mechanical measurements are therefore the primary evidence and the diffraction is corroboration. What ex-situ diffraction cannot do is observe the material while loaded, and an in-situ measurement is named as the direct test.

Comment 7.

Response. We respectfully push back rather than half-comply. The purpose of this work is to test whether an LLM-hypothesized composition has the potential for stress-induced martensitic transformation; the Omori composition is here as a control, not a subject. Its superelasticity — including the dependence on grain size, orientation, texture and B2 state the Reviewer lists — is among the most thoroughly documented subjects in this family's literature (Omori et al., Science 333 (2011) 68), and this paper has nothing to add to it. What the control must do is validate the methods, and Fig. 3b is that demonstration: 0.5% recovery on heating under the same protocol that resolves nothing in the LLM-alloy. A fully corresponding stepwise characterization of the benchmark would re-measure settled literature without sharpening the paper's question, and we have deliberately not added one.

On corresponding characterization more broadly, the revision reports what corresponds and is candid about what cannot. Both alloys' 1200 °C grain structures are characterized and both responses to the AGG treatment are in Sec. 3.5; the LLM-alloy's texture is quantified (Fig. 7) while for the benchmark the same coarse structure that defeats Rietveld quantification precludes an orientation-distribution refinement, which Sec. 3.3 states along with its cause. Where symmetry is impossible, Sec. 3.3 explains why it does not affect the comparison the paper makes: what the coarse structure withholds is a number for how much of the benchmark transformed, not the fact that it transformed.

Comment 8.

Response. The logical point — absence of a peak does not prove absence of the phase — is correct, and the revision responds in three ways. Every categorical “no B2” statement is softened to what the measurement supports: no B2 superlattice reflection was resolved above the detection limit of the present measurement. That limit is then quantified rather than asserted, which converts the concession into a bound (Sec. 3.3): because B2 and D0<sub>3</sub>

share the {200}-type position, the discriminating measurement is the  $\text{D}_{03}$ -only {111} reflection, and any B2 in the LLM-alloy is at least four times less abundant than in the benchmark and may be absent altogether — with the text conceding that a finer or more dilute dispersion cannot be excluded by diffraction, and naming TEM or atom-probe tomography as what would be required. Third, the equilibrium analysis answers the heat-treatment question at a more fundamental level than peak detection: coherent B2 forms by ageing a quenched single-phase  $\alpha$  matrix, and this alloy has no such field at any accessible temperature, which also explains the total absence of response to the 200 °C/3 h age.

Comment 9.

Response. On the two cast heats alone we cannot — the Reviewer is right and the submitted manuscript overclaimed. The revision answers with two calculations that change one variable at a time, which no pair of physical heats can do. The first is a virtual carbon-free control: deleting  $\approx 0.10$  wt% carbon from the measured chemistry, everything else held fixed, moves the  $\alpha$  solvus from  $\approx 1340$  to  $\approx 1150$  °C and converts the 1200 °C equilibrium from duplex to 100%  $\alpha$  in all three databases (Fig. 9c). The second addresses the Reviewer's specific candidate — a nickel scan from 4.2 to 7.8 at%, across which the solvus does not move at all, so restoring the benchmark's nickel would not open the window by a degree; what nickel governs instead is how much ordered bcc can form. Sec. 4.2 states the separation in those terms: nickel sets how much B2 could form, carbon sets whether there is ever an  $\alpha$  matrix in which to form it coherently.

Comment 10.

Response. Accepted in full. Sec. 4.1 now reads that coherent B2 precipitates “contribute to, rather than solely determine, transformation reversibility,” followed by the Reviewer's list stated as such, with both supplied references cited; the revision adds La Roca et al.'s point that by drawing Ni and Al out of solution the precipitates also shift the matrix's transformation temperatures, so ageing moves both the obstacle field and the thermodynamic setting at once. We note that the revision's own diagnosis rests on the first item of that list — parent-phase stability — rather than on B2, with the precipitation question strictly downstream.

Comment 11.

Response. Accepted. Micrographs were not taken at every anneal temperature, so the recovery/recrystallization/growth reading is an inference from the mechanical response, and Sec. 4.3 now presents it as exactly that: the interpretive language is downgraded throughout, the text states explicitly that the threshold is inferred rather than observed, and EBSD across the anneal series is named as the experiment that would settle it. Both quoted phrases are gone — “fully recrystallized” no longer appears as a definitive claim, and “single-phase-controlled structural metal” is deleted outright, the Reviewer having been doubly right there since this alloy is explicitly two-phase and the phrase was wrong on its face.

Comment 12.

Response. We concede the overgeneralization and have corrected it, and ask the Reviewer's consideration for retaining one bounded methodological claim. The sweeping language is gone: statements of the form “AI models cannot yet capture...” are replaced by findings about this case, Sec. 4.4 carries an explicit scope paragraph stating that one composition proposed by one agent in one session and processed by one route is a case study rather than an evaluation of AI-assisted alloy design, and the first limitation in the Conclusions says the same. The revision also strengthens the case-study half on its own

terms, as the framing invites, with the alloy ultimately classified against the lightweight-steel family whose chemistry it shares.

What we ask to retain is one methodological conclusion that survives the  $n = 1$  objection because it does not rest on the sample. The failure mode found here — a required phase field lying above the temperature range in which the alloy can be processed — was diagnosed by an equilibrium calculation that takes minutes, uses openly available databases, and is validated in this same study against the benchmark. “LLM-proposed compositions should be screened this way before synthesis” is a conclusion about a workflow step: its warrant is the diagnosis, not a statistic over candidates, and it would remain true even if most LLM-proposed alloys succeeded, because screening is cheap and failure is expensive at any failure rate. Sec. 4.4 grounds it in a documentary finding rather than a generalization — the agent’s own report recommends the CALPHAD screening that was never performed — so the lesson attaches to where the workflow terminated, not to what the model knew.

We thank the Reviewer again for their comments and time regarding our manuscript.

Reviewer 3 - Revision 1, 04/09/2026

The authors have revised the manuscript in response to the previous round of comments, but several issues remain that directly affect the manuscript’s core conclusions regarding the absence of super-elasticity and the underlying material mechanisms:

1. The revised manuscript has substantially changed the core conclusions and scientific interpretation of the original manuscript, going beyond ordinary supplementation and refinement. Although revising conclusions in light of new evidence is a normal scientific process, such a marked change also indicates that the original manuscript drew overly definitive conclusions on the basis of insufficient evidence, raising concerns about the rigor of the scientific argument. The authors should clearly state whether the original conclusions still hold and demonstrate that the revised core conclusions are supported by sufficiently reliable experimental evidence.

2. The three limitations listed by the authors in the Conclusions actually further weaken the current core conclusions regarding AI-guided alloy design and the scientific contribution of the manuscript. The authors have acknowledged that this study involves only one candidate composition, one LLM session, and one processing route; whether a different processing window could enable phase transformation in this composition has not been tested; and the carbon-free control has only been virtually evaluated by calculation. More importantly, the authors argue that a key reason for the present failure is that the AI did not determine “whether the temperature required for the desired phase stability could actually be reached under the available equipment and processing conditions,” and they point out that a simple equilibrium phase-diagram pre-screening could have excluded this candidate before melting. If the problem arises mainly because the design workflow did not incorporate phase-diagram information, processing temperature, and equipment constraints, rather than because AI itself is unable to propose a candidate with super-elasticity, then what this study actually demonstrates is closer to an incomplete AI prompting and subsequent human-screening workflow, rather than a new scientific limitation of AI-guided alloy design itself. In fact, the manuscript does not test whether explicitly including the single-phase  $\alpha$  field, transformation temperatures, CALPHAD results, and actual processing temperatures at the prompting and screening stages would yield a different or even viable candidate. The authors therefore need to clarify what the actual scientific contribution of this study is. If the core contribution is only that “AI-proposed candidates should undergo CALPHAD and processing-window screening before experiments,” the authors should explain the novelty of this conclusion



relative to conventional alloy-design workflows.

3. The cyclic heat-treatment route in Section 2.2 requires further clarification. The Methods state that the specimen was held at 1200 °C, water quenched, and then reheated to 900 °C. The authors should clarify whether this was the actual experimental route and whether it is consistent with the Omori-type cyclic heat treatment in Ref. 18 that traverses the  $\alpha/\alpha+\gamma$  phase fields. Since the revised manuscript argues that the LLM-alloy cannot obtain a single-phase  $\alpha$  parent phase at 1200 °C, the authors should also explain why a cyclic heat-treatment schedule close to that of the benchmark alloy was still adopted and what this treatment was expected to achieve in the LLM-alloy.

4. There is a clear inconsistency in the gauge length reported for the mechanical testing in Section 2.4. The original manuscript stated 13 mm, whereas the revised manuscript changes this to 127 mm and states that strain was calculated as crosshead extension/127 mm. The two values differ by nearly a factor of ten, yet Fig. 3 still uses the original curves. The authors must clarify whether 13 mm was a typographical error in the original manuscript, what gauge length was actually used for each figure, and which tests were repeated or which data were reprocessed. This issue directly affects the reliability of the strain, recoverable strain, and related conclusions.

5. English: Fig. 3a tests the LLM-alloy after treatment at 1200 °C  $\times$  1 min. At this point, the material is already a  $\gamma$ -majority FCC/BCC duplex microstructure, and the fraction of BCC parent phase available for the stress-induced  $\alpha \rightarrow \gamma$  transformation is clearly insufficient. Therefore, Fig. 3a can only demonstrate, in the first instance, that this specific microstructural state does not exhibit obvious super-elasticity; it cannot directly prove that the composition itself is incapable of exhibiting super-elasticity. The relevant conclusions should be strictly limited to the microstructural and processing conditions that were actually tested in this study.

6. In Section 3.4, the authors rely mainly on CALPHAD calculations to conclude that the single-phase  $\alpha$  field of the LLM-alloy lies above approximately 1340 °C, while also citing an incipient melting temperature of approximately 1240 °C, and on this basis conclude that the single-phase  $\alpha$  state is experimentally inaccessible. However, the former is an equilibrium thermodynamic prediction, whereas the latter is derived mainly from a single database, and different databases do not give consistent solidus predictions. These temperatures therefore should not be treated directly as established material properties. High-temperature DSC/DTA is recommended to verify the incipient melting temperature.

7. In Section 3.5, the evidence that the 1200 °C  $\times$  40 min condition and the AGG condition show “no phase transformation” remains insufficient. The authors report only that the elongation of the condition corresponding to Fig. 1c decreases to 1.4%, but do not present the complete cyclic stress–strain curves, loading–unloading–heating curves, recoverable strain or residual strain data for this condition, nor do they quantitatively compare the  $\alpha/\gamma$  phase fractions in Fig. 1a and Fig. 1c. The authors further use the absence of transformation in the 1 min and 40 min conditions to infer that the AGG condition would likewise show no transformation, but the AGG condition itself was not subjected to the same cyclic super-elasticity testing. The authors are advised to provide directly the complete cyclic mechanical results and phase-constitution data for both the 1200 °C  $\times$  40 min condition and the AGG condition, rather than using the results from two conventionally annealed conditions as a substitute for direct experimental evidence from the AGG condition.

8. Section 4.1 proposes that the  $\alpha \rightarrow \gamma$  transformation may already have occurred during cooling and further speculates that  $M_s$  may be above room temperature. Since this explanation has become an important mechanism used to account for the  $\gamma$ -majority phase constitution at room temperature and the absence of super-elasticity,  $M_s$  should be measured directly.